

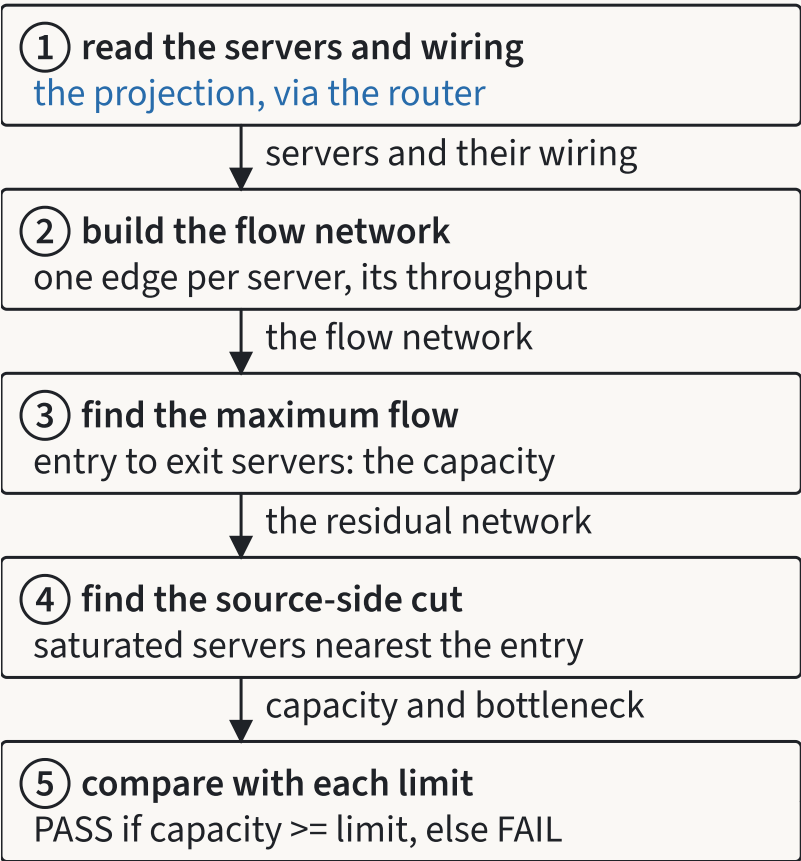
# L2b Pipeline example: capacity and verdicts

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status: approved without external review, 2026-08-27

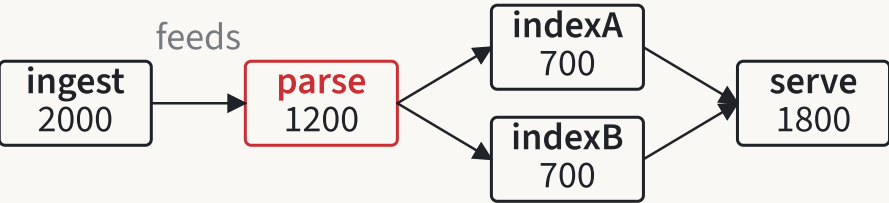
Why does raising one server change nothing and raising another change everything?  
the capacity service and the example model, after the demo has been run once

## functional view: how the number is made

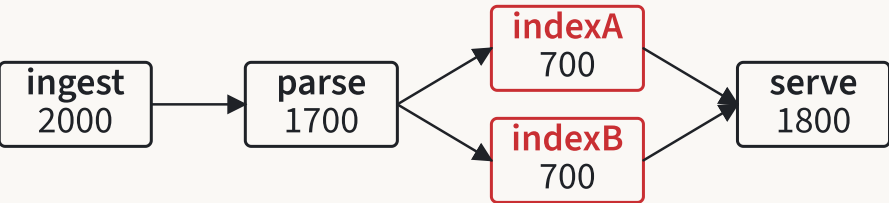


known issue: limits of validity  
exact for the idealised model,  
an upper bound for a real one that  
conserves flow, and not for  
capacity planning. The assumptions:  
flow is conserved at every server  
even partitioning across branches  
perfect load balancing  
no queueing, no latency coupling  
stationary load, no bursts  
unlimited connections

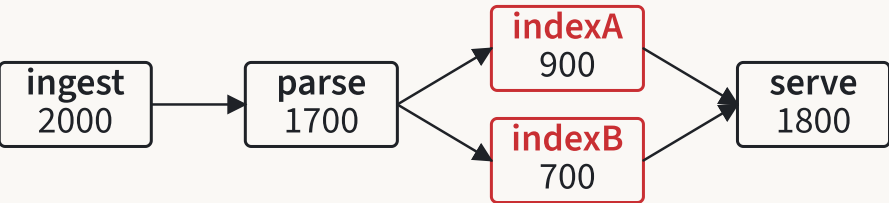
the wiring, in queries per second  
shipped: 1200, PIPE-R1 **FAIL**



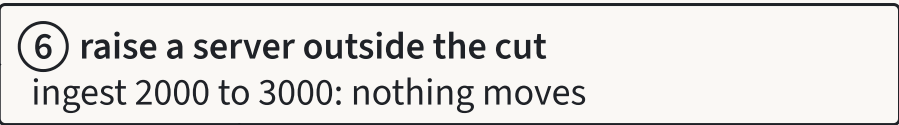
parse to 1700: 1400, PIPE-R1 **FAIL**



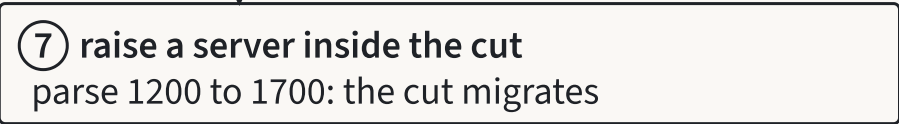
then indexA to 900: 1600, PIPE-R1 **PASS**



verdicts and reasons



the same number, the same cut



then indexA 700 to 900: capacity 1600,  
PIPE-R1 **PASS**, and PIPE-R1.4 on indexB  
still fails, 700 against its allocated 750

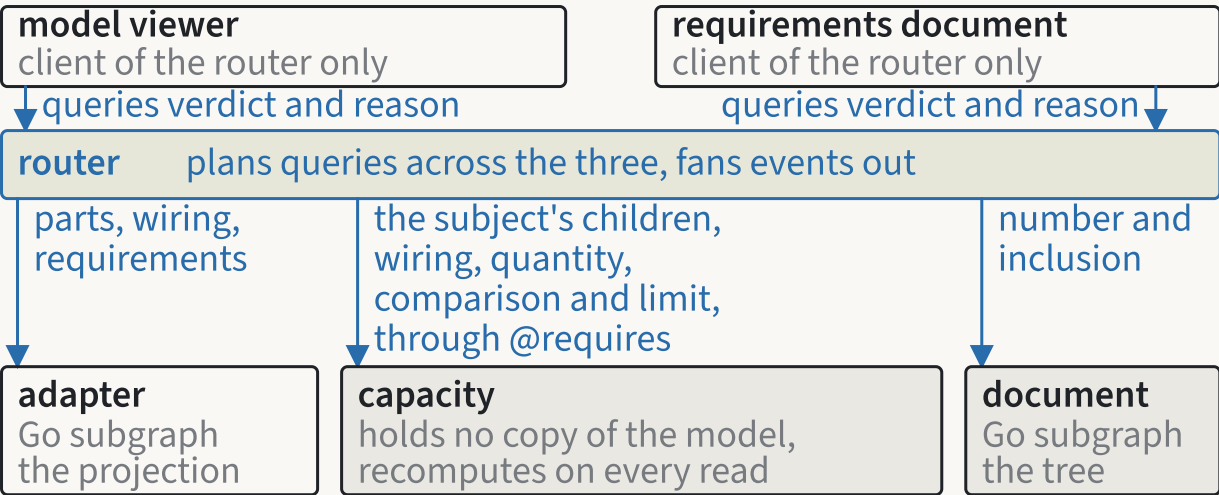
## the arithmetic

capacity = the maximum flow, entry to exit servers, each  
server one edge sized by its throughput = the minimum cut:  
min over a chain, sum over parallel branches

| state       | cut           | capacity, PIPE-R1 | derived                                                          |
|-------------|---------------|-------------------|------------------------------------------------------------------|
| shipped     | parse         | 1200 <b>FAIL</b>  | PASS <b>FAIL</b> <b>FAIL</b> <b>FAIL</b> <b>PASS</b>             |
| ingest 3000 | parse         | 1200 <b>FAIL</b>  | PASS <b>FAIL</b> <b>FAIL</b> <b>FAIL</b> <b>PASS</b>             |
| parse 1700  | indexA indexB | 1400 <b>FAIL</b>  | PASS <b>PASS</b> <b>FAIL</b> <b>FAIL</b> <b>PASS</b>             |
| indexA 900  | indexA indexB | 1600 <b>PASS</b>  | PASS <b>PASS</b> <b>PASS</b> <b>PASS</b> <b>FAIL</b> <b>PASS</b> |

derived, in order: ingest, parse, indexA, indexB, serve  
limits: 1500 on the pipeline, ingest, parse and serve,  
750 on each index server. Row 2 is box 6, row 4 is box 7:  
PIPE-R1.4 fails at 700 against its allocated 750 while  
PIPE-R1 passes  
reasons capacity 1200 against 1500, limited by parse  
capacity 1400 against 1500, limited by indexA, indexB  
capacity 1600 against 1500, limited by indexA, indexB  
all values known, from the running example,  
guarded by a unit test

## where it runs: three services, one router



## decisions

- ★ AD-0006 idealised model  
only throughput limits a server

★ AD-0024 reasons from fixed templates  
a reason carries no word of the model  
the record: docs/architecture and docs/decisions, the capacity model in the example README.  
Sibling sheet: L0. Decision records: AD-0006, AD-0007, AD-0024, AD-0025.
- ★ AD-0007 maximum flow, source-side cut  
any wiring, the nearest cut is unique

★ AD-0025 the document owns its structure  
it owns order, nesting and numbering

- a component

□ owned by the example

□ the router and what  
passes through it

□ a failing verdict or  
the bottleneck

★ a decision record, its id  
and one line of rationale  
(the cookbook types stars  
by concern)

① the reading path  
1200 a known value